

FTO gene

Numerous studies show that FTO variants are linked with an average BMI that is 1.0 to 3.0 kg/m² higher.

Recent research shows that FTO is involved in regulating signaling molecules that are involved in appetite and weight regulation in the hypothalamus.[\[ref\]](#)[\[ref\]](#)

- Early studies link FTO and BMI to increased energy intake due to eating more fat.[\[ref\]](#) But that hasn't been replicated.
- People with variants in the FTO gene express more FTO, possibly causing higher **ghrelin** ('hunger hormone') levels.[\[ref\]](#)
- Animal studies show that increased FTO causes a **decreased amplitude of the core circadian rhythm genes**.[\[ref\]](#) Disruption of the core circadian genes is linked to obesity in a lot of studies.

Check your genetic data for **rs9939609** (23andMe v4, v5; AncestryDNA)

- A/A: higher risk of obesity, increased BMI, increased FTO expression[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)[\[ref\]](#) even greater impact in women with PCOS (average increase of over 20 lbs)[\[ref\]](#)
- A/T: increased risk of obesity, increased BMI, increased FTO expression
- T/T: typical

Members: Your genotype for rs9939609 is TT.

Lifefacks for FTO:

- **Time-restricted eating:** The FTO gene may impact circadian rhythm. Look into time-restricted eating — essentially, eating your meals during certain hours of the day and then not eating the rest of the evening/night. Try giving your body a break from eating for more than 12 hours each night.
- **Get more active:** People with the highest levels of physical activity don't seem to have an impact from the FTO gene on their BMI.[\[ref\]](#) Even being 'lightly active' is better than being sedentary for people with the FTO risk alleles.[\[ref\]](#)

Related article: [FTO and Weight](#)

MC4R gene

Genetic variants that decrease MC4R levels are linked to increased BMI. MC4R is important in how the brain receives the signal from leptin, which is the satiety hormone. Without that 'stop eating' satiety signal, people can have an increased appetite driving them to eat a little more than what their body needs.

- MC4R deficiency is not only tied to increased appetite and higher BMI throughout life but also linked to greater muscle mass as well.[\[ref\]](#)[\[ref\]](#)
- A study looked at infants carrying the MC4R variant and found no differences in birth weight based on the variant, but by two weeks of age, there is a difference in BMI and appetite.[\[ref\]](#)

Check your genetic data for **rs17782313** (23andme v4, v5):

- C/C: risk of increased BMI in both children and adults[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)
- C/T: risk of increased BMI in both children and adults
- T/T: typical

Members: Your genotype for rs17782313 is **CT**.

Check your genetic data for **rs17700633** (23andme v4, v5; AncestryDNA):

- A/A: risk of increased BMI, obesity[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)
- A/G: risk of increased BMI, obesity
- G/G: typical

Members: Your genotype for rs17700633 is **AG**.

Lifehacks for MC4R:

- **Protein:** One study found that those with an MC4R deficient variant had better weight loss results on a lower protein diet compared to a high protein diet (with no difference found for carbs and fat variation).[\[ref\]](#)
- **Stay Active:** In studies of MC4R genetic variants in childhood obesity, more active children with the variant were less likely to be obese than sedentary children with the variant.[\[ref\]](#)

Related article: [Growing up big boned: MC4R variants](#)

NPY gene

Neuropeptide Y (NPY) is released in the brain due to stress. It is abundantly found in the brain, including in the hypothalamus, where it regulates appetite. Increasing NPY in the brains of lab animals causes a many-fold increase in food consumption for the next 6 – 8 hours.[\[ref\]](#)

Check your genetic data for **rs16139** (23andMe v4, v5; AncestryDNA):

- T/T: typical
- C/T: higher BMI, higher leptin levels, increased food intake
- C/C: (rare) higher BMI, higher leptin levels[\[ref\]](#)[\[ref\]](#)

Members: Your genotype for rs16139 is TT.

Lifehacks for NPY:

- **Watch out for stress-eating carbs:** NPY overactivation is linked to increased consumption of carbs and sweets. If you tend to stress eat, try limiting the amount of carbohydrates or sweet snacks available.
- **Consistency if fasting:** Animal studies show that short-term fasting causes an increase in NPY expression and rebound 'hyperphagia' (desire to eat).[\[ref\]](#) But longer-term fasting, or consistent intermittent fasting, may still lead to weight loss.[\[ref\]](#)

LEPR Gene

The LEPR gene codes for the leptin receptor. Leptin is the satiety signal that turns off your drive to eat. Most of the genetic variants below are very common; in fact, up to half of some populations carry the risk variant.

Check your genetic data for **rs1137101** Q223R (23andMe v4, v5 ; AncestryDNA):

- A/A: typical
- A/G: increased risk of overweight /obesity; diabetes; PCOS;
- G/G: increased risk of overweight /obesity[\[ref\]](#); diabetes[\[ref\]](#); PCOS[\[ref\]](#); weight gain on antipsychotic meds[\[ref\]](#); greater muscle gain with weight training[\[ref\]](#);

Members: Your genotype for rs1137101 is **AG**.

Check your genetic data for **rs1137100** K109R (23andMe v4, v5; AncestryDNA):

- A/A: typical
- A/G: somewhat increased risk of obesity, diabetes
- G/G: increased risk of obesity, diabetes[\[ref\]](#)[\[ref\]](#)

Members: Your genotype for rs1137100 is —.

Lifehacks for LEPR:

- **Reduce Omega 6 Fats:** People with LEPR who eat a diet low in omega-3 and high in omega-6 are at a significantly increased risk of metabolic syndrome. Adding more omega-3 fats (such as fish) and decreasing omega-6 fats (such as fried foods) may help.[\[ref\]](#)[\[ref\]](#)
- **Low-carb diet to reduce leptin:** Cutting down on carbs (e.g., 20% or less of your calories from carbohydrates) has been shown to decrease leptin.[\[ref\]](#)
- **Increase Melatonin:** Leptin has a daily rhythm impacted by sleep and light. Melatonin, which is decreased by light at night, plays a key role in leptin levels. [\[ref\]](#)[\[ref\]](#)[\[ref\]](#) Increase melatonin at night by blocking out blue light or through supplementation.
- **Quality Sleep:** A recent study showed that people who slept 5 hours or less at night had a 15% decrease in leptin levels and a 14% increase in ghrelin.[\[ref\]](#)

SH2B1 gene

SH2B1 codes for a signaling protein that increases the response to leptin in the brain. Without as much SH2B1, there is a decrease in leptin signaling, resulting in increased appetite and weight gain.[\[ref\]](#)[\[ref\]](#)

Check your genetic data for **rs7359397** (23andMe v5; AncestryDNA):

- C/C: typical
- C/T: higher BMI
- T/T: higher BMI, increased risk of obesity[\[ref\]](#)[\[ref\]](#)

Members: Your genotype for rs7359397 is **CT**.

Lifefacks for SH2B1:

- **Watch your fat intake.** Animal studies clearly show that SH2B1 is downregulated by eating a diet high in fat.[\[ref\]](#) Studies in humans show that the SH2B1 variants are linked to obesity with high dairy intake. In studies, a 'high dairy intake' indicates an increased amount of saturated fat in the diet (think butter and cheese instead of skim milk).[\[ref\]](#)[\[ref\]](#)

CLOCK gene

The aptly named CLOCK gene codes for a key part of the core molecular circadian clock that governs your body's rhythms over the course of 24 hours. Eating later at night is linked in numerous studies to increased weight gain.[\[ref\]](#)[\[ref\]](#)

Check your genetic data for **rs1801260** (23andMe v4, v5; AncestryDNA):

- G/G: higher activity in the evening, delayed sleep onset, increased risk for obesity, harder to lose weight[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)
- A/G: somewhat increased risk for obesity
- A/A: typical

Members: Your genotype for rs1801260 is **GA**.

Lifefacks for CLOCK:

- **Cut out the evening snacking:** Realize that you may tend to stay up later and want to snack in the evening. Large studies show that people who eat one or two meals a day generally weigh less than people who eat 3-6 meals/day. A benefit was also shown for having a longer overnight fast.[\[ref\]](#)
- **Mediterranean diet - meh:** No beneficial effects were noted for people with the G-allele when eating a Mediterranean-style diet for a year. This was compared to people carrying the CLOCK rs1801260 A/A genotype, who had significant benefits in insulin sensitivity with the diet.[\[ref\]](#)
- **Get enough sleep:** Not getting enough quality sleep each night is linked to weight gain. People who tend to stay up a little later may cut short their sleep each morning in order to go to work or school. Adjusting your bedtime and

making quality sleep a priority may help to prevent weight gain in CLOCK variant carriers.[\[ref\]](#)

Related article: [CLOCK: Genes at the core of our circadian rhythm](#)

SLC6A14 (Serotonin) gene 🔥

The SLC6A14 gene codes for a transporter that regulates how tryptophan gets used for serotonin synthesis. Serotonin is a neurotransmitter important in appetite regulation and fatty acid utilization.[\[ref\]](#) SLC6A14 is located on the X-chromosome, so women will have two copies of the gene, and men will have one.

Check your genetic data for **rs2011162** (23andMe v4 only)

- G/G: decreased fatty acid oxidation (use of fat for energy) in women, increased risk of obesity[\[ref\]](#)
- C/G: decreased fatty acid oxidation
- C/C: typical

Members: Your genotype for rs2011162 is —.

Lifehacks:

- 5-HTP (serotonin) is a supplement that is often advertised for fat loss. While it may work for some people, it is likely to decrease the use of fat for energy even more in SLC6A14 variant carriers.[\[ref\]](#)

UCP1 Gene 🔥

The UCPI gene codes for Uncoupling Protein 1 found in the mitochondria of brown fat tissue. Brown fat is the 'good' fat that burns a lot of energy, producing more heat. People with higher levels of brown fat are more likely to be lean.

Check your genetic data for **rs1800592** (23andMe v4, v5; AncestryDNA):

- T/T: typical
- C/T: slightly increased risk for obesity
- C/C: weak UCP1 activity, higher risk of abdominal fat, generally higher risk of obesity[\[ref\]](#)[\[ref\]](#)

Members: Your genotype for rs1800592 is —.

Lifehacks for UCP1:

Cold exposure: One way to stimulate brown fat is by exposing yourself to the cold. Things to try include:

- Cold showers, or a shower that ends with 30 seconds of cold water.
- Turn down the heat in the winter.
- Ice baths or cold water plunges (make sure you know what you are doing here!)

Caution: Being cold activates brown fat through norepinephrine release from the adrenal glands. Therefore, if you are already stressed out, have poor thyroid function, or have adrenal fatigue, using cold to induce brown fat may add stress to your already stressed system.

Related article: [Turning up the internal heat for weight loss: UCP1](#)

Conclusion:

Understanding how your genes specifically relate to your weight can help you decide on a weight loss plan.

Picking a Plan:

In general, sticking to an eating plan that fits your lifestyle works best in the long run. The main premise of most diet plans is to cut out highly processed foods, avoid soda, and eliminate fast food. Depending on your genetic data above, you may want to steer away from a lot of fat in your diet and instead include more vegetables. Others may find a higher-fat, lower-carb diet helpful for regulating appetite.

Here are a few resources to get you started:

- **Time-Restricted Eating:** To learn more about the science behind time-restricted eating, the book *"The Circadian Code"* by [Dr. Panda](#) is a great place to start.

- **Paleo Diet:** The 'Paleo' diet focuses on eating mainly vegetables, fruits, and meat – leaving out grains and dairy. Learn more at thepaleodiet.com.
- **Mediterranean Diet:** Often promoted for heart health, a Mediterranean-style diet focuses on fresh vegetables, fruits, whole grains, and seafood. Learn more and find recipes on themediterraneandish.com.
- **Low-Carb diet:** Cutting down on carbs (e.g., 20% or less of your calories from carbohydrates) has been shown to decrease ghrelin and leptin.[\[ref\]](#) Google keto recipes or low-carb diet.

Recap of your genes:

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FTO	rs9939609	TT	A	0.39	Increased BMI; increased risk of obesity (variant with the largest impact, circadian rhythm and appetite)
MC4R	rs17782313	CT	C	0.23	Increased BMI; increased muscle mass; higher appetite
MC4R	rs17700633	AG	A	0.27	Increased BMI; increased muscle mass; higher appetite
NPY	rs16139	TT	C	0.03	Increased BMI; higher leptin levels, increased food intake (hunger driven by brain)
LEPR	rs1137101	AG	G	0.45	Increased risk of being overweight or obese; weight gain on antipsychotic medications; greater muscle gain with weight training
LEPR	rs1137100	--	G	0.26	Increased risk of being overweight or obese; increased diabetes risk
SH2B1	rs7359397	CT	T	0.35	Higher BMI, increased risk of obesity, increased response to leptin
CLOCK	rs1801260	GA	G	0.26	Increased risk for obesity, harder to lose weight, evening chronotype, core circadian clock gene
SLC6A14	rs2011162	--	G	0.52	Decreased fatty acid oxidation; increased risk for obesity; serotonin synthesis gene
UCP1	rs1800592	--	C	0.32	Weak UCP1 activity, higher risk of abdominal fat, generally higher risk of obesity